BUILD A CUBESAT

BAC Guide Rules – Condensed Checklist for Reviewing Utils

CONTENTS

Project-Wide	2
CLI Tools (Section 3)	3
Terminal Presentation (Interface Guide Section 10)	3
Marimo Notebooks (Section 4)	4
Mission Profile Contract (Section 2.7.1)	5
Web and HTML Surfaces (Interface Guide Sections 3, 4, 5, 11)	5
Blender Extensions (Section 5)	6
Shell Scripts (not explicitly covered – apply the spirit)	6

- Source: BAC Project & Tooling Guide v0.16 (2026-10-29), BAC Interface Design Guide v1.1 (2026-10-29), BAC Identity Guide v1.4 (2026-10-30) and BAC Trademark Policy v1.2 (2026-10-29).

Project-Wide

- License: software MIT; pyproject.toml has license = { text = "MIT" }. Hardware CERN OHL-S 2.0, mechanical parts and bac-structure included. Docs CC BY-SA 4.0, except READMEs and instructions that belong to a design, which follow its repository. Copyright line: Copyright (c) <years> nihil.space GmbH and Build a CubeSat contributors on the open Codeberg repositories; the GitHub repositories keep their nihil.space GmbH line; bac-ops (private) is all rights reserved.
- Legal entity: nihil.space GmbH only; never the LLC.
- Brand marks (name, BAC, cube mark, wordmark, lockup) are not openly licensed: BAC Trademark Policy. A repository that ships logo artwork says so in its README's license section.
- Software versioning: SemVer, displayed with a v prefix (bac-toolname v0.1.0). Hardware: v<version>r<revision>[.<patch>][-<variant>]. Git tags <tool>-v<X.Y.Z> only where another repo pins the version; merge commits (git merge --no-ff), never squash, never rebase a tagged branch.
- Attribution fields: legal-entity field → nihil.space GmbH; project field → Build a CubeSat; one mixed field → Build a CubeSat by nihil.space GmbH. KiCad Company, STEP organization, 3MF Designer are entity fields; STL has none.
- Naming: "Build a CubeSat" on first reference, "BAC" thereafter, never Bac or BaC. Lowercase bac- in code and filenames; never mixed case.
- Typography: Gedankenstrich (–) only, never em dash (—). Straight apostrophes. ' thousands separator for CHF and metric, comma for USD. × not x. Decimal point. Headings numbered 1., 1.1 – no trailing period on subsections.
- Headings, titles and labels are Title Case everywhere (lowercase articles, conjunctions, short prepositions). American spelling everywhere: color, gray, behavior, license.
- The thousands separator is display only. CSV, JSON and TOML carry plain unformatted numbers.
- Axis directions use letters: Xm, Xp, Ym, Yp, Zm, Zp. Never -X or +Y. Lowercase in command-line values and file names (--face zp).
- Titles are matter-of-fact: "Power Budget Breakdown", not "Where the Watts Go".
- No emojis. Only semantic glyphs: ✓ pass, × fail, ! warning, → progression, · or , separator.
- Status labels (Section 2.11): `DEVELOPMENT`, `TESTING`, `RELEASED`, `DEPRECATED`, `DO NOT USE`, `ARCHIVED`, `EXTERNAL` – uppercase words as inline code, each mapped to an Interface Guide state; never an emoji.
- Data formats: TOML for hand-edited config, CSV for tabular, JSON machine-to-machine, avoid YAML except where a framework or service requires it (Zephyr, F´, mkdocs.yml, workflow files).
- File naming for data files: {project}-{descriptor}-{YYYY-MM-DD}.{ext}; no version numbers in filenames.
- Design tokens: tokens.json in bac-utils (tools/bac-design-tokens) is the executable source of the Identity Guide's tokens, MIT or CC BY-SA 4.0. bac-design-tokens writes the gallery's and bac-suite's tokens.css, the Rich colors in bac_common.ui and every marimo style cell; never edit a generated copy, run uv run --no-sync bac-design-tokens and commit; --check fails CI on a stale copy. Other repos (BAC Ground) take a copy of the generated tokens.css.
- Repos: Codeberg primary (codeberg.org/buildacubesat-project/...); GitHub only for bac-page, CubeSat-Resources, bac-utils and bac-design (the guides).
- Time: UTC; Unix epoch machine-readable, ISO 8601 human-readable.
- Every doc about a versioned thing has a version history section. READMEs and user guides follow their subject's version, with no separate doc version; BAC-FSW-* and BAC-SBC-* documents keep their own versions; docs-site pages without a versioned subject take their history from git (last-updated date).
- SKUs: bac-<cat>-<sub>-<name>-<versionstring>[-<variant>], validated before API use.

CLI Tools (Section 3)

- Python 3.11+, uv, pyproject.toml compatible with uv tool install, hatchling build backend, package bac_toolname, script bac-toolname = "bac_toolname.cli:main", minimum-pinned deps with no upper bounds.
- Stack: Rich for output; argparse for simple tools, Typer for multi-subcommand; InquirerPy for guided prompts (4 or more choices, or costly mistakes; plain [y/N] fine for low stakes); tomllib for config; python-dotenv for secrets.
- Standard flags: -v/--version and hidden --debug on every tool; the rest where they apply. -v/--version (non-negotiable, prints toolname v0.x.x), -l/--list where a named set exists, --env-file PATH, --dry-run (prints what would happen, labeled "dry run – no changes made"), --debug (hidden from --help, shows tracebacks), --init (first-time setup; the tool detects missing setup and points to it), --config PATH for tools with persistent settings (default ~/.config/bac/{toolname}.toml).
- Secrets: never hardcoded, .env plus python-dotenv, commit .env.example without values, .env in .gitignore.
- Filters write to stdout unless -o FILE; batch tools skip an existing output with a ! line and exit 0, --force redoes it; two inputs to one output is an error before any work.
- In-place editing tools: plan every change, show it, ask once (--yes for scripts, --dry-run stops before), keep <file>.bak unless --no-backup, edit by span so the diff shows only the change, leave a file that changed under the tool alone and report it.
- Errors: stderr, raise BacError or a subclass (UsageError exit 2, ConfigError, ExternalToolError) and let bac_common.cli.run print it, no tracebacks unless --debug, exit 0/1/2. Rich rendering: [#E94538]ERROR[/] Short description. plus an indented suggestion.
- --help is 24 lines or fewer at 80 columns (48 for a subcommand whose options are the tool's vocabulary), counted on Python 3.11; one-sentence description, defaults noted, 2 or 3 examples; Typer examples in the epilog behind a \b line.
- Reference implementation: bac-common in bac-utils (make_parser/run_typer, run, ui, config, testing), lib/bac-common/tests/demo_tool.py as the shape of a tool.

Terminal Presentation (Interface Guide Section 10)

- Rich colors through bac_common.ui's generated constants: success [#3FB950]✓, error [#E94538]×, warning [#F7D400]!, labels [bold], muted [dim], paths and identifiers [cyan]. One hex per role, tuned for dark terminals. At most three active colors per screen state; one dominant yellow.
- Liveness: anything silent for more than about 1 second shows a spinner or progress bar (rich.progress). Progress bar: task name, bar (#F7D400), one right-aligned status (x/n, or %, or elapsed). Collapse progress after completion.
- Tables: rich.table.Table, default HEAVY_HEAD; box.SIMPLE for secondary tables; numerics right-aligned; do not wrap cells.
- Panels only for the opening run summary and the final result block (Panel(..., box=SIMPLE, border_style="dim")). Rules for section dividers. Opening panel: [bold]bac-toolname[/bold] [dim]v0.1.0[/dim] \nOne-sentence description. Canonical header: Build a CubeSat – toolname v0.1.0.
- Run structure: opening panel → step log (one print per step, ✓/× lines) → result summary (numbers, always shown even on partial failure; dry run appends [dim](dry run – no changes made)[/dim]).
- Destructive or costly actions require explicit confirmation or before/after review.
- Terminal tools do not implement light and dark themes.

Marimo Notebooks (Section 4)

- Stack: marimo via uv, pandas or polars, Altair primary, Matplotlib secondary, Plotly for 3D only, and pyarrow>=15.0 in every notebook that shows a table.
- WebAssembly notebooks (molab, mkdocs-marimo on cubesat-resources.space): Pyodide's package set at Pyodide's versions only; no compiled packages outside its list, sgp4 among them; PEP 723 metadata uses minimum bounds, never ==; molab does not rewrite those bounds; 2 GB memory cap.
- Header cell is a mo.vstack of a three-paragraph intro (what it estimates and how to drive it, what to compare, links) and mo.callout(kind="warn", title="Planning Estimates Only"). No "!" intro line.
- Cell order: header, style, imports, constants, shipped profiles, profile loader (upload / parse / status, three cells), control elements, panel layout, sidebar, panel render, resolve, computation, chart conventions, headline cards, model callouts, map, charts, tables, export, assumptions and acknowledgment, revision history.
- Two constraints force the splits: a cell cannot read a UI element it defines, and mo.sidebar must be the last expression of its cell. Dataflow, not file position, sets execution order, so the export cell may read ASSUMPTIONS_MD defined below it.
- Style cell is the canonical shared text, written by bac-design-tokens (never edited by hand; notebook-specific rules go after its last line): light-dark() rather than a prefers-color-scheme media query, --heading-font set or headings fall back to Lora, --bg light-dark(#F9F9F9, #1F1E1D), --text light-dark(#3F3F3F, #E5E5E5), --text-muted light-dark(#6D6D6D, #A9A8A6), Nunito Sans for headings (500) and body, IBM Plex Mono for code only, 10px callouts, no .vega-embed rule (charts render on canvas in a shadow root).
- Headings and callout titles are Title Case with lowercase articles, conjunctions and prepositions, as everywhere. Notebook copy uses American spelling.
- UI elements: show_value=True on every slider, plus include_input=True where typed entry helps; numeric controls built through clamping N()/S() helpers, since an out-of-range value takes the panel cell down; data_editor values validated (finite numbers, tolerant _flag() for checkboxes, duplicate names skipped and reported); a control's value is unreadable in its defining cell; preset-versus-profile resolved by comparing the preset's current value with what the profile said; sidebar switch pattern; two-column panel default with accordions for rarely-touched groups; mo.ui.number shows a comma separator and cannot be overridden.
- Charts: alt.theme.enable(...), not the deprecated alt.themes.enable; width='container' does not cross alt.vconcat; downsample below Altair's 5'000-row transformer cap; a log axis dies on a 0 in the series; chart conventions live in one shared cell; the chart title carries the legend semantics; axis titles always explicit. Chart text, axes, ticks and subtitles carry no color (marimo's chart theme sets them; labelColor, titleColor, domainColor, tickColor, subtitleColor are banned by a test). PALETTE is the shared notebook palette ["#2E7ED8", "#7D7D7D", "#9D7800", "#0E903A", "#E4463E"] and MUTED = "#A9A8A6" if IS_DARK else "#6D6D6D", both generated.
- Maps: keyed CARTO tiles as mark_image with .scale(None), mercator scale = 256·2^z/(2π), clip=True on every layer, fixed 960 × 540 px, zoom slider, tiles-off switch with the Natural Earth coastline as fallback, legends orient="bottom", chart rendered bare because mo.ui.altair_chart refuses geoshapes, credit line underneath, ground tracks split on |Δlon| > 180.
- Output: six gray mo.stat(bordered=True) cards under "Headline Numbers", no direction= coloring – the callouts do the judging. Callout kinds: warn model limit or failed check, danger hard impossibility, info note, success the single all-clear. Cross-tool callouts named "X Against Y", shown only when the sibling's results table is present.
- Exports: labels "Download report (.md)", "Download profile (.toml)", "Download … (.csv)"; names bac-<tool>-<qualifier>-<date>.<ext>; report header Generated YYYY-MM-DD HH:MM UTC (Unix N) with BAC <Tool> <version>, bac.page/<slug>. Profile: <name>.; CSV never runs through fmt_int.
- Assumptions text: bold-lead paragraphs, one per model area, a Limitations paragraph, one closing line; glossary deep links on first use to https://cubesat-resources.space/references/glossary/#slug; no bullet lists.

- Tooling: marimo check --fix before every commit, marimo check as the CI lint step; headless tests via app.run() and tests/harness.py; marimo export html --no-include-code as the cheapest check that every Altair spec validates. Headless checks miss empty charts (dotted field names, layers filtered to nothing) – render specs with vl-convert once per chart change; Vega has no apostrophe thousands, use labelExpr.
- Repository layout: notebooks/bac-<name>/ with bac_<name>.py, README.md, tests/, examples/, a pyproject.toml whose version equals TOOL_VERSION (test enforced); SPDX line first; no bac-common (not in Pyodide); exempt from -v/--help; ruff with E501, B018, F841 off. Two bac.page links per notebook, molab-<tool>-app and molab-<tool>, both reading main; a moved number is a new version with a revision row and re-recorded figures.
- Compiled models (pymsis) are evaluated offline by a PEP 723 script beside the notebook and embedded as a text table; never edited by hand; a test compares against a fresh evaluation.

Mission Profile Contract (Section 2.7.1)

- Top level: name, tool (bac_<tool>), tool_version.
- [spacecraft] (body, mass, drag coefficient, drag attitude as an axis: ram_x, tumbling...) and [[deployables]] are written by the lifetime notebook and read by thermal, orbit viewer and power budget; pointing directions go in the owning tool's table in letter form (nadir_zm); panels stay in the power budget's [[panels]], matched to deployables by name.
- Normative shared tables: [orbit] (altitude_km, inclination_deg, ltdn_hours, epoch, sim_days, min_elevation_deg), [ground_station] (station, latitude_deg, longitude_deg, min_elevation_deg), [target] (name, latitude_deg, longitude_deg), [map] (tiles, zoom, key).
- Results in [results.<tool>] at the end, per-row results in [[results.<tool>.modes]], ignored when the writing tool reloads its own file.
- A missing key keeps the loading tool's default; a tool ignores what it does not own instead of failing.
- Author detected from tool, else from a signature table ([[modes]], [sensor], [storage]). The loader status block states what was taken and what was ignored.
- Writer is hand-rolled because Pyodide has no tomli_w: escape backslash, quote, newline, carriage return and tab; unwrap numpy scalars; drop NaN and infinity; lowercase booleans; header comment as one block.
- Reader is P() with dotted sections, bool coercion, float-from-integer-default, and a profile_warnings list; PH() distinguishes absent from empty.

Web and HTML Surfaces (Interface Guide Sections 3, 4, 5, 11)

- prefers-color-scheme by default, light when the system reports no preference (CSS cannot tell that case from light), with an optional [data-theme] override; test both. Inside marimo, follow marimo's own setting with light-dark() instead.
- WCAG AA contrast (--text-muted #6D6D6D/#A9A8A6, --link #0B5CAD/#4C9AFF, status inks on their tints), focus is a 3 px solid --focus-ring outline (#0B5CAD/#4C9AFF) with 2 px offset and no halo, prefers-reduced-motion respected, color never the only state channel – every state is color + shape + word.
- Motion tokens: ease-standard cubic-bezier(0.2,0,0.1,1); durations 120/200/320 ms; no bounces or infinite loops.
- States: nominal green #3FB950 filled circle, caution yellow #F7D400 triangle, alarm red #E94538 square, info blue #004C8B diamond, no data/planned gray dashed hollow circle; text uses the per-theme ink (green #1A7F37/#3FB950, red #C9302C/#FF5A4F, info #0B5CAD/#4C9AFF, caution #7A5F00/#F7D400). No functional amber; #4277FD is retired, blue means info, links and focus.

- Surfaces: --bg #F9F9F9 light, #1F1E1D dark; cards #FFFFFF/#282726; wells lighter than the page in dark. Dark mode has no accent: gray primary buttons, neutral --accent-state; yellow means caution only. Light primary buttons are yellow with dark text.
- Form: pill buttons, chips and tabs with uppercase IBM Plex Mono labels at 0.03em; 6px inputs, 10px cards; no shadows, no hover lifts. Body Nunito Sans 400, headings Nunito Sans 500, every number in mono tabular figures.
- Charts: ops charts use the console plot colors (--chart-1..6, per theme; no --chart-4/-5 as plain series where the chart carries pass/fail); Mission charts the nebula triad (teal #0E9594, violet #7A5AF8, rose #C64677, max three series); notebooks the shared palette.
- Icons: Lucide, inline SVG (no CDN script), 18px in icon buttons; status shapes are not icons. Containers 1200px (reading) and 1540px (dashboards). At most one info-blue hero card per view; its red rule is decorative.
- Ops versus Mission mode from the Identity Guide; controls in one region; labels on every input; units next to values.

Blender Extensions (Section 5)

- VSE tools: bl_category = "VSE Tools"; non-VSE: bl_category = "BAC".
- UI labels Title Case, as everywhere. No abbreviations unless standard Blender, such as FPS and AVI.
- Three sub-panels: Bulk Import, Sequential Proxies, Strip Utilities. Sub-panel order is fixed: one-line layout.label description → full-width operator buttons with a built-in icon and text → property rows.
- Property names snake_case; display names Title Case with units in parentheses ("Pause (s)"); description= required on every property, a complete sentence, no trailing period.
- Destructive or long-running operators need a confirm dialog or a visible before/after indicator; long-running ones are modal timers (PASS_THROUGH, cancel(), settings in a scene property group), invoked with "INVOKE_DEFAULT".

Shell Scripts (not explicitly covered – apply the spirit)

- Same flags where meaningful (-v/--version, --dry-run, -h), fail loudly (set -euo pipefail), errors to stderr, exit codes 0/1/2, no emojis, Gedankenstrich, version history in the header or README.

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Source: guides/bac-project-and-tooling-guide-condensed.md · exported by bac-design-export 0.1.0
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